The documentation and process conversion measures necessary to comply with this revision shall be completed by 12 March 1998.

INCH POUND

MIL-PRF-19500/349D
12 December 1997
SUPERSEDING
MIL-S-19500/349C
3 December 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPES 2N3506, 2N3506A, 2N3506L, 2N3506AL, 2N3507, 2N3507L, 2N3507A, 2N3507AL,
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to T0-39).

1.3 Maximum ratings. 1/

P_T 2/ T_A = +25°C	P_T 3/ T_C = +25°C	V_{CBO}		V_{CEO}		V_{EBO}	I_C	T_J and T_{STG}
W	W	2N3506 2N3506A	2N3507 2N3507A	2N3506 2N3506A	2N3507 2N3507A	V dc	A dc	°C
1.0	5.0	V dc 60	V dc 80	V dc 40	V dc 50	5.0	3.0	-65 to +200

1/ Electrical characteristics for "A", "AL", and "L" suffix devices are identical to non "L" suffix devices unless otherwise noted.
2/ Derate linearly 5.71 mW/°C above T_A = +25°C.
3/ Derate linearly 28.6 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics. 1/

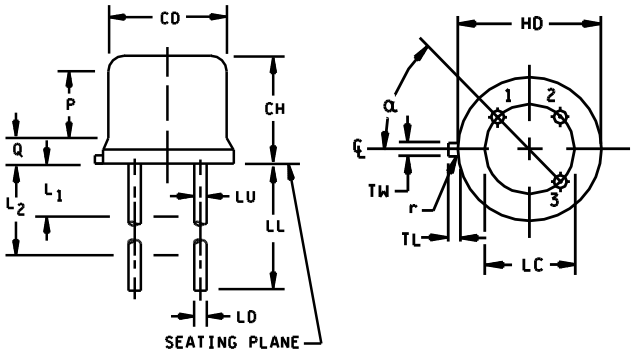
	h_{FE2} 2/		h_{FE4} 2/		$V_{CE(sat)}$	$\|h_{fe}\|$	Cobo	t_{on}	t_{off}
Limits	V_{CE} = 2.0 V dc I_C = 1.5 A dc		V_{CE} = 5.0 V dc I_C = 3.0 A dc		I_C = 1.5 A dc I_B = 150 mA dc	f = 20 Mhz V_{CE} = 5 V dc I_C = 100 mA dc	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	I_C = 1.5 A dc I_B = 150 mA dc	
	2N3506 2N3506A	2N3507 2N3507A	2N3506 2N3506A	2N3507 2N3507A					
					V dc		pF	ns	ns
Min	40	30	25	20		3			
Max	200	150			1.0	15	40	45	90

1/ Electrical characteristics for "A", "AL", and "L" suffix devices are identical to non "L" suffix devices unless otherwise noted.
2/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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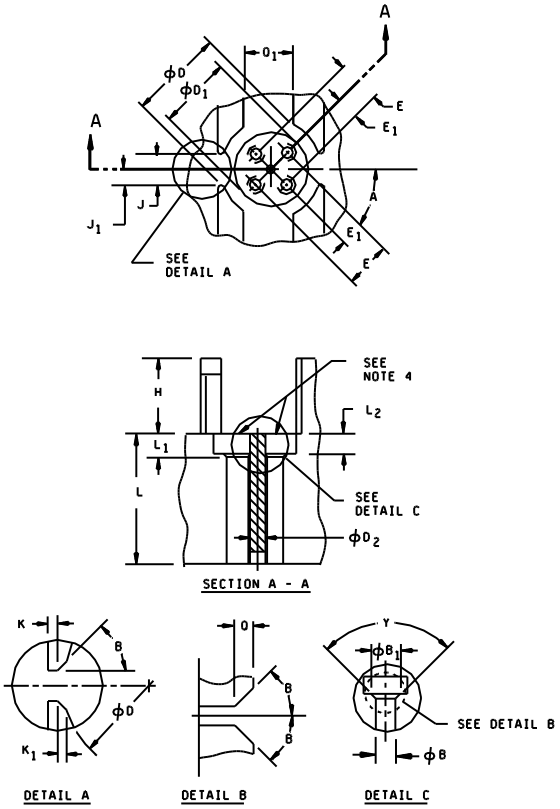
Dimensions					
Symbol	Inches		Millimeters		Notes
CD	0.305	0.335	7.75	8.51	
CH	0.240	0.260	6.10	6.60	
HD	0.335	0.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	0.016	0.021	0.41	0.53	7, 8
LL	See notes				7, 8, 11,12
LU	0.016	0.019	0.41	0.48	7, 8
L1		0.050		1.27	7, 8
L2	0.250		6.35		7, 8
P	0.100		2.54		5
Q		0.050		1.27	4
r		0.010		0.25	10
TL	0.029	0.045	0.74	1.14	3
TW	0.028	0.034	0.71	0.86	2
α	45° TP		45° TP		6

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. For 2N3506L, 2N3507L, 2N3506AL, and 2N3507AL, dimension LL shall be 1.5 inches (38.1) mm minimum and 1.75 inches (44.4 mm) maximum.
12. For 2N3506, 2N3506A, 2N3507, and 2N3507A dimension LL shall be 0.5 inches (12.7) mm minimum and 0.75 inches (19.0 mm) maximum.
13. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
14. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (similar to TO-39).

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Ltr	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
ϕB	.8255	.8509	.0325	.0335	5
ϕB_1	1.09 Nom		.040 Nom		5
ϕD	5.867	5.880	.2310	.2315	
ϕD_1	4.04	4.09	.159	.161	
ϕD_2	1.02 Nom		.040 Nom		6
E	2.527	2.553	.0995	.1005	
E_1	1.257	1.283	.0495	.0505	
H	3.68	3.94	.145	.155	
J	1.194	1.207	.0470	.0475	
J_1	.597	.622	.0235	.0245	
K	.229	.279	.009	.011	
K_1	.127 Nom		.005 Nom		
L	9.45	9.60	.372	.378	
L_1	1.37	1.40	.054	.055	
L_2	1.09 Nom		.043 Nom		
Q	1.02 Nom		.040 Nom		
Q_1	3.12	3.23	.123	.127	
α	---	---	44.90°	45.10°	
B	---		45° Nom		
Y	---		90° Nom		



NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) ±.010 inch (0.254 mm) from the seating surface of the gauge. A force of 8 ±.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
4. These surfaces to be parallel and in same plane within ±.001 inch (0.025 mm).
5. Four holes.
6. Pressed in.

FIGURE 2. Gauge for lead and tab location.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (similar to T0-39).

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

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4.3 Screening. Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
7	hermetic seal (optional)	1/
9	h_{FE2}, I_{CEX1}	N/A
10	48 hours minimum	48 hours minimum
11	I_{CEX1}; h_{FE2}; ΔI_{CEX1} = 100% of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15% of initial value	I_{CEX1} and h_{FE2}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX1} = 100% of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15% of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100% of initial value or 200 nA dc, whichever is greater; Δh_{FE2} = 15% of initial value.

1/ Hermetic seal test shall be performed in either screen 7 or screen 14.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = +25° C ± 3° C; V_{CB} = 30 V dc; P_T = 1.0 W. JANS devices - - V_{CB} = 10 V dc.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal impedance ($Z_{\Theta JX}$ measurements). The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current - - - - - - - - - - - 10 mA.
b. I_H forward heating current - - - - - - - - 500 mA (min).
c. t_H heating time - - - - - - - - - - - - - 25 - 30 ms.
d. t_{md} measurement delay time - - - - - - - - 60 µs max.
e. V_{CE} collector-emitter voltage - - - - - - - 10 V dc minimum

The maximum limit for $Z_{\Theta JX}$ under these test conditions are $Z_{\Theta JX}$ (max) = 17.5° C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2.

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4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	2037	Test condition A
B4	1037	V_{CB} = 10 V dc; T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750; P_T = 1.0 W; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted.
B5	1027	V_{CB} = 10 V dc; T_A = +125° C ± 25° C for 96 hours, P_T = 1.0 W at T_A = +100° C or adjusted as required according to the chosen T_A to give an average T_J = +275° C.
B6	3131	See 4.5.2

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	HTRB, test condition A, T_A = 150° C, V_{CB} = 80% of para 1.3 rated, 340 hours, n = 22 devices, c = 0.
2	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 V dc, P_T = 1 W; at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted. n= 45 devices, Acc = 0
	or	
	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq$ +100° C; forced air cooling allowed on cooling cycle only. n = 45 devices, Acc = 0.
3	2037	Bond strength, test condition E, n = 11 wires, c = 0.
4	1032	High-Temperature life (non-operating), T_A = +200° C n = 22, c = 0

1/ Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Samples shall be selected from each wafer from a die fabrication line that the manufacturer chooses to "certify" for the die design used in products manufactured to this specification.

b. Must be chosen from an inspection lot that has been submitted to and passed group A conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B3 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.2.4 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die meeting the following requirements:

a. Manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

b. The die lot was manufactured (using start or completion dates, as the manufacturer selects to consistently use) within three months following the wafer lot used for sample selection in paragraph 4.4.2.3. Group B shall not be required more than once for any single wafer lot.

4.4.3 Group C inspection, JANS. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 herein (JANS). See 4.4.3.2 for JAN, JANTX, and JANTXV group C testing. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2.

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4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	conditions: 1,000 hours at V_{CB} = 10 V dc; P_T = 1 W, at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on device shall be permitted.

4.4.3.2 Group C inspection, JAN, JANTX, and JANTXV. Group C inspection (JAN, JANTX, and JANTXV, package related tests) shall be as follows and shall include those package or case-related tests which are performed periodically. Group C test on each package family shall be performed on devices from each six months production of devices (based on inspection lot identification codes) for each assembly location and die attach method. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2. The sample size for each of these subgroups is: 22 devices, c = 0. Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

Step	Method	Condition
1	2066	
2	1051	Test condition C.
3	2036	Test condition E.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.3.4 Product acceptable for delivery. The product which may be accepted for delivery based a successful completion of group C testing shall be product of the particular package type with lot inspection identification codes of the 52 successive weeks beginning with the inspection lot identification code of the successful group C sample (i.e. the inspection lot identification codes accepted for delivery based on successful group C testing will include those sealed during the approximate 10 weeks during which the sample was being screened and tested plus those sealed in the 42 weeks following successful completion of group C testing making a total of 52 successive calendar weeks.) A manufacturer shall not accept product for delivery until after successful completion of the group C testing on product of the applicable package type with an inspection lot identification code less than 51 weeks prior to the inspection identification code of the product being considered for acceptance.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Minimum collector current magnitude during power application shall be 160 mA dc for $R_{\Theta JC}$ and 32 mA dc for $R_{\Theta JA}$.

b. Collector to base voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case for $R_{\Theta JC}$ and ambient air for $R_{\Theta JA}$.

d. Reference point temperature shall be selected with $25^\circ \leq T_R \leq 35^\circ$ C and recorded before test is started.

e. Mounting arrangement shall be with heat sink to case for $R_{\Theta JC}$ and without heat sink for $R_{\Theta JA}$.

f. Maximum limits shall be $R_{\Theta JC} = 35^\circ$ C/W and $R_{\Theta JA} = 175^\circ$ C/W.

NOTE: Unless otherwise specified by the acquiring activity, the choice for measuring $R_{\Theta JC}$ or $R_{\Theta JA}$ will satisfy the requirements of this test.

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TABLE I. Group A inspection.

Inspection 1/, 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Solderability 1/ 2/	2026	15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	15 devices, c = 0				
Subgroup 2						
Breakdown voltage collector to base	3001	Bias condition D, I_C = 100 µA dc	$V_{(BR)CBO}$			
2N3506				60		V dc
2N3507				80		V dc
Breakdown voltage emitter to base	3026	Bias condition D, I_E = 10 µA dc	$V_{(BR)EBO}$	5		V dc
Breakdown voltage collector to emitter	3011	Bias condition D, I_C = 10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3506				40		V dc
2N3507				50		V dc
Collector to emitter cutoff current	3041	Bias condition A, V_{EB} = 4 V dc	I_{CEX1}		1	µA dc
2N3506		V_{CE} = 40 V dc				
2N3507		V_{CE} = 60 V dc				
Forward-current transfer ratio	3076	V_{CE} = 1 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE1}			
2N3506				50	250	
2N3507				35	175	
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 1.5 A dc, pulsed (see 4.5.1)	h_{FE2}			
2N3506				40	200	
2N3507				30	150	
Forward-current transfer ratio	3076	V_{CE} = 3 V dc, I_C = 2.5 A dc, pulsed (see 4.5.1)	h_{FE3}			
2N3506				30		
2N3507				25		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 3.0 A dc, pulsed (see 4.5.1)	h_{FE4}			
2N3506				25		
2N3507				20		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/, 2/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limit Min	Limit Max	Unit
Subgroup 2 - Continued						
Saturation voltage and resistance	3071	I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.5	V dc
Saturation voltage and resistance	3071	I_C = 1.5 mA dc, I_B = 150 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.0	V dc
Saturation voltage and resistance	3071	I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)3}$		1.5	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 500 mA dc, I_B = 50 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.0	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 1.5 A dc, I_B = 150 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$	0.9	1.4	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)3}$		2.0	V dc
Subgroup 3						
High-temperature operation		T_A = +150° C				
Collector to emitter cutoff current	3041	Bias condition A, V_{EB} = 4 V dc	I_{CEX2}		1.5	mA dc
2N3506		V_{CE} = 40 V dc				
2N3507		V_{CE} = 60 V dc				
Low-temperature operation		T_A = -55° C				
Forward-current transfer ratio	3076	I_C = 500 mA dc,pulsed (see 4.5.1)	h_{FE5}			
2N3506, 2N3506L		V_{CE} = 1.0 V dc		25		
2N3507, 2N3507L		V_{CE} = 1.0 V dc		17		
2N3506A, 2N3506AL		V_{CE} = 2.0 V dc		25		
2N3507A, 2N3507AL		V_{CE} = 2.0 V dc		17		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/, 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc, I_C = 100 mA dc, f = 20 MHz	$\|h_{fe}\|$	3	15	
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		40	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 3 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		300	pF
Pulse response:						
Delay time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = 150 mA dc, (see figure 3)	t_d		15	ns
Rise time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = 150 mA dc, (see figure 3)	t_r		30	ns
Storage time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = I_{B2} = 150 mA dc, (see figure 4)	t_S		55	ns
Fall time	3251	Test condition A, I_C = 1.5 A dc, I_{B1} = I_{B2} = 150 mA dc, (see figure 4)	t_f		35	ns

1/ For sampling plan see MIL-PRF-19500.
2/ Electrical characteristics for "A", "AL", and "L" suffix devices are identical to non suffix devices unless otherwise noted.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ Not required for JANS devices.

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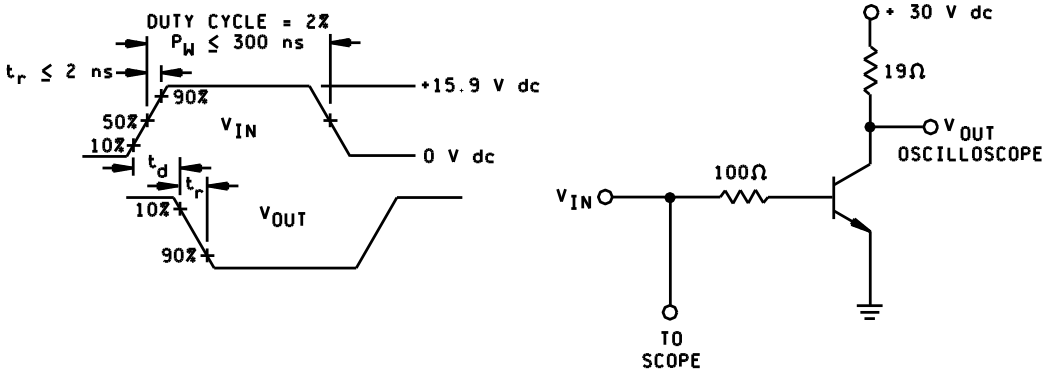


FIGURE 3. Equivalent circuit for measuring delay and rise times.

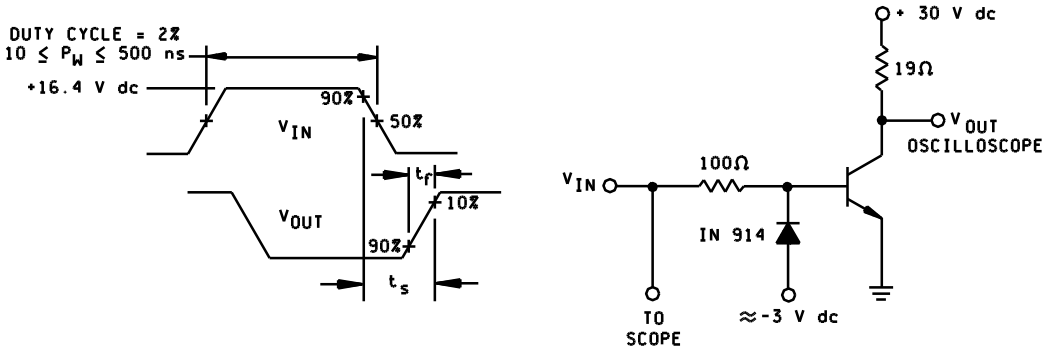


FIGURE 4. Equivalent circuit for measuring storage and fall times.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead finish as specified (see 3.3.1).

b. Type designation and product assurance level.

6.3 Substitution information. The 2N3506A and 2N3507A may be substituted for the 2N3506 and 2N3507 (and L suffix versions) (respectively) at the discretion of the procuring activity."

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, MI
Navy - AS, CG, MC
Air Force - 13, 19, 85
NASA - NA

Preparing activity:
DLA - CC

(Project 5961-1933)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/349D	**2. DOCUMENT DATE (YYMMDD)** 97/12/12

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING TYPES 2N3506, 2N3506A, 2N3506L, 2N3506AL, 2N3507, 2N3507L, 2N3507A, 2N3507AL, JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE, FAX (Include Area Code), E-mail: (1) Commercial DSN FAX E-mail 614-692-0510 850-0510 614-692-6939 alan_barone@dscc.dla.mil
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

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